

Title:

Business Intelligence Implementation & Dashboard Development

E-Commerce Customer Behavior & Sales Analysis

1.0 Introduction:

Business Intelligence (BI) is the practice of using technologies, tools, and processes to collect, process, analyze, and present data in a way that helps organizations make informed decisions. It transforms raw data from multiple sources into meaningful insights that guide strategy, improve operational efficiency, and support real-world analytical decision-making.

1.1 BI System:

BI systems consist of several key components that work together to convert data into actionable insights. First, data sources refer to the origin of data, such as the Kaggle e-commerce dataset, which contains 5,000 transactions from a Turkish online retail platform (January 2023 to March 2024), including customer demographics, purchasing behavior, and product preferences.

1.2 ETL (Extract, Transform, Load):

ETL is the process of extracting data from sources, transforming it into a clean and consistent format, and loading it into a centralized system. For example, extracting raw data from Kaggle, converting it into Excel format, and loading it into Power BI for analysis.

1.3 Data Warehouse:

The data warehouse is a centralized repository where structured and processed data is stored, enabling efficient querying, historical analysis, and reporting based on the assignment dataset.

1.4 Dashboards and Reports:

Dashboards and reports present data visually through charts, graphs, and key performance indicators (KPIs), making it easier to interpret insights. For instance, three dashboards have

been created in Power BI to analyze sales performance dashboard, customer analysis dashboard and financial overview trends.

1.5 BI Architecture

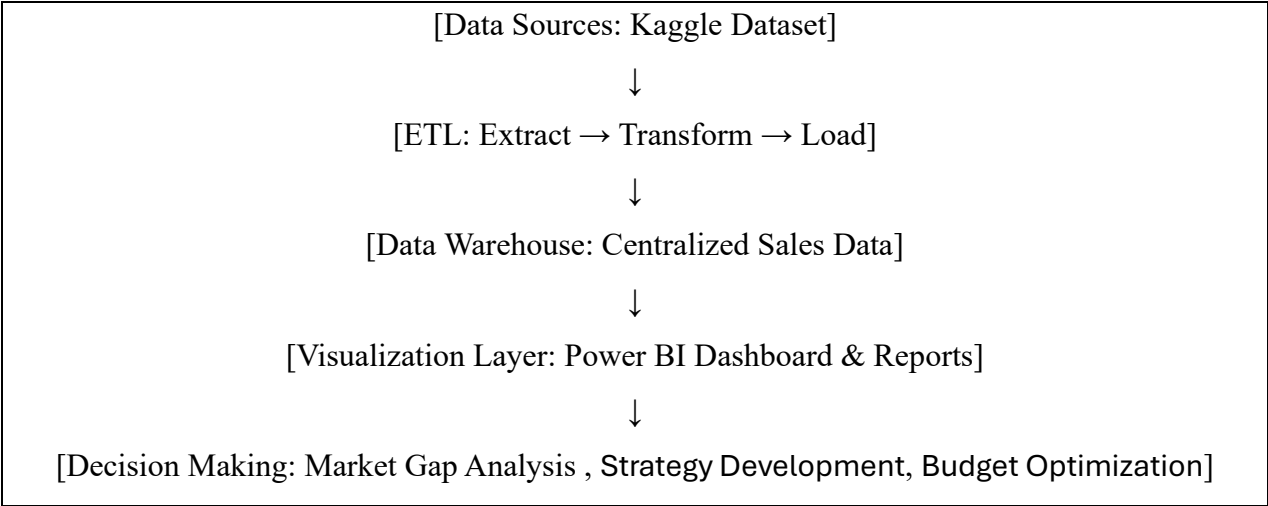


Table 1: BI Architecture Design with description of layers Decision-Making Support

The BI Architecture for this project was designed to minimize latency between data acquisition and strategic action. By implementing a robust ETL layer, we ensured that the resulting visualizations such as the Sales Performance and Customer Analysis dashboards provide a verified baseline for identifying operational risks and planning resource distribution across the e-commerce landscape

Task 2: Dataset Selection & Description

2.0 For this task, the selected dataset is the “E-Commerce Customer Behavior & Sales Analysis-TR, sourced from Kaggle. This dataset contains 5,000 real-world e-commerce transactions from a Turkish online retail platform, spanning from January 2023 to March 2024. It provides detailed information that helps analyze customer behavior, purchasing patterns, product categories, and transaction trends, making it suitable for business intelligence and analytics projects.

The dataset includes **18 columns** capturing various attributes related to orders, customers, products, and customer interaction metrics.

Key variables include **Order_ID** (unique identifier for each order), **Customer_ID** (unique user identifier), **Age** and **Gender** (demographics), **City** (customer location), **Product_Category** (eight product categories such as Electronics, Fashion, Home & Garden, etc.), **Unit_Price** and **Quantity** (pricing and quantity purchased), **Discount_Amount** and **Total_Amount** (transaction values after discount), and **Payment_Method** (Credit Card, Bank Transfer).

The dataset also contains behavior metrics such as **Device_Type**(used to make the purchase. Options: Mobile, Desktop, Tablet) , **Session_Duration_Minutes**,(Time spent on website during the session in minutes. Range: 1-120 minutes) **Pages_Viewed**,(Number of pages viewed during the shopping session. Range: 1-50 pages) **Is_Returning_Customer**,(Whether the customer has made previous purchases. Values: True (returning) or False (new customer)) **Delivery_Time_Days**,(Number of days taken to deliver the order. Range: 1-30 days) and **Customer_Rating**, which provide insights into customer engagement and satisfaction, Customer satisfaction rating for the order. Scale: 1-5 stars (1=very dissatisfied, 5=very satisfied))

Task 3: Data Preprocessing & Transformation - Perform the following: Data cleaning (handling missing values, duplicates) Data transformation (formatting, aggregation) Explain ETL process briefly Tools allowed: Excel, Power BI, Tableau, Python

3.0 The ETL Process Briefly Explained

ETL stands for Extract, Transform, and Load. It is the backbone of any Business Intelligence system, ensuring that raw data is turned into "clean" information for the dashboard.

- **Extract:** imported the raw CSV dataset ("E-Commerce Customer Behavior & Sales ") from Kaggle into the Power BI Desktop environment using the Get Data feature.
- **Transform:** Used the Power Query Editor to clean the data. This involved changing data types (e.g., setting Sales to Currency), removing any null values in the 'Gender' column, and creating a Date Hierarchy to allow for Year-over-Year (YoY) analysis.
- **Load:** Once the data was cleaned and structured, I loaded it into the Power BI Data Model to begin building the visualizations.

3.1 Data Cleaning Steps Performed

To ensure the accuracy of the **Sales Performance** and **Financial** dashboards, I performed the following:

- **Handling Missing Values:** scanned the Customer_ID and Total_Sales columns for nulls. Any incomplete transaction records were removed to prevent skewed KPIs. Check for nulls in "Gender" or "Age." If minor, remove those rows; if significant, impute with the "Average."
- **Removing Duplicates:** "Remove Duplicates" feature on the Order_ID column to ensure that the Total Orders count was a true reflection of unique transactions.
- **Correcting Data Types:** Ensured that the Age and Quantity columns were set as Whole Numbers, while Total_Sales and Total_Profit were set to Decimal Number (Currency) for proper summation.

Column Name	Correct Data Type
Order_ID	Text
Customer_ID	Text
Date	Date/Time
Age	Whole Number
Gender	Text
City	Text
Product_Category	Text
Unit_Price	Decimal Number
Quantity	Whole Number
Sales	Decimal Number
Discount_Amount	Decimal Number
Total_Amount	Decimal Number
Payment_Method	Text
Device_Type	Text
Session_Duration_Minutes	Decimal Number
Pages_Viewed	Whole Number
Is_Returning_Customer	Text/Boolean
Delivery_Time_Days	Whole Number
Customer_Rating	Decimal Number

3.2 Data Transformation & Aggregation

- **Formatting:** Formatted the Date column to follow a standard DD/MM/YYYY format so the line charts would display chronological trends correctly.
- **Measures Creation:** Created custom DAX Measures for complex calculations that weren't in the raw data, such as:
 - **Average Order Delivery Time (AOD):** Calculated as $\$Total\ Sales \div Total\ Orders\$$.
 - **Total Profit %:** Calculated as $\$Total\ Profit \div Total\ Sales\$$.
- **Categorization:** Grouped the Product Category data to enable the Tree Map and Waterfall visuals, allowing for a high-level view of which departments drive the most revenue.

Task 4: Dashboard Development -Develop an interactive dashboard

4.0 Introduction

The E-Commerce Customer Behavior and Sales Analysis dashboard is a comprehensive business intelligence tool designed to transform raw transactional data into actionable strategic insights. In the modern retail landscape, the ability to investigate the intersection of consumer demographics and financial performance is critical for sustained growth. This dashboard serves as a central "command center," providing a multi-dimensional view of organizational health through high-fidelity data visualizations and interactive analytical engines.

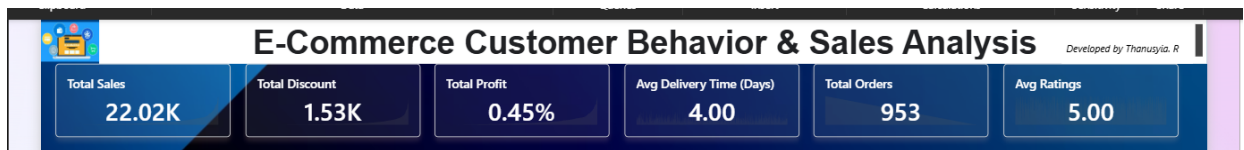
Tools : Microsoft POWERBI



Figure 1: This report is segmented into specialized modules, including a Financial Overview for auditing profitability, a Customer Analysis for demographic segmentation, and a Forecast engine for predictive planning.

4.1 Dashboard 1: KPI Dashboard

Key Performance Indicators (KPIs)

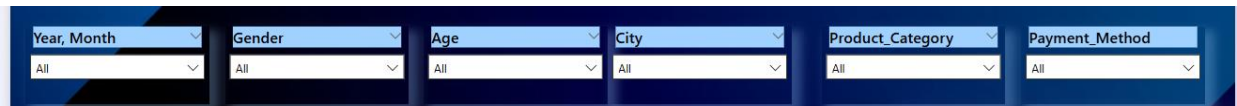


The top row acts as the "Heartbeat" of the business, featuring six distinct pillars of performance:

- **Total Sales (\$22.02K):** The primary growth metric representing gross revenue.
- **Total Discount (\$1.53K):** Visibility into promotional impact.

- **Total Profit (0.45%):** A critical financial health indicator showing the efficiency of the bottom line.
- **Avg. Delivery Time (4 Days):** A logistics metric tracking the speed of fulfillment.
- **Total Orders (953):** A volume indicator representing market activity.
- **Avg. Ratings (5.00):** A direct pulse on customer satisfaction and product quality.

Slicer Functional Breakdown



- **Temporal & Geographic:** **Year, Month,** and **City** filters allow for precise time-series analysis and regional performance auditing.
- **Customer Persona:** **Gender** and **Age** slicers isolate specific demographics to uncover unique purchasing habits and preferences.
- **Operational & Financial:** **Product Category** and **Payment Method** filters identify high-value departments and the most profitable transaction channels.

4.2 Sales Performance Dashboard

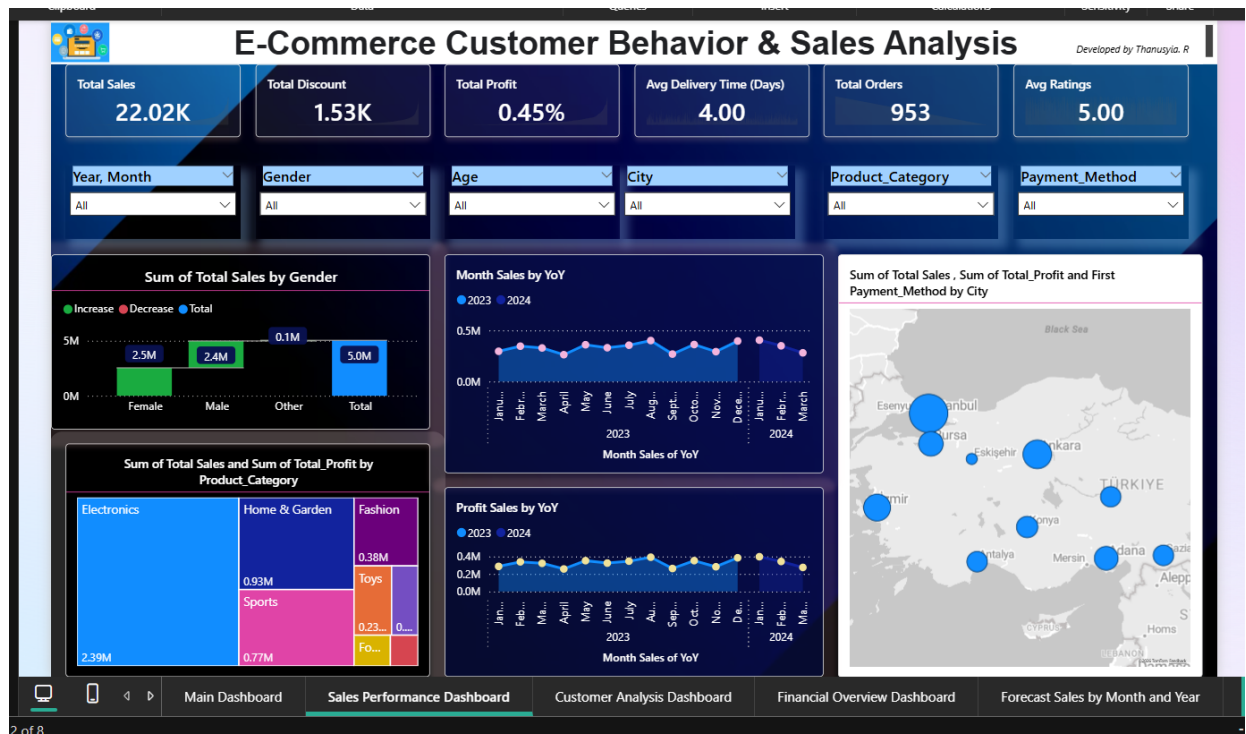


Figure 2: Presents the Sales Performance Dashboard, an advanced analytical interface designed to provide a 360-degree view of the business's operational and geographic footprint. This "command center" serves as the primary gateway for investigating how various market dimensions such as geography, gender, and product category intersect to drive the platform's \$5.0M total revenue.

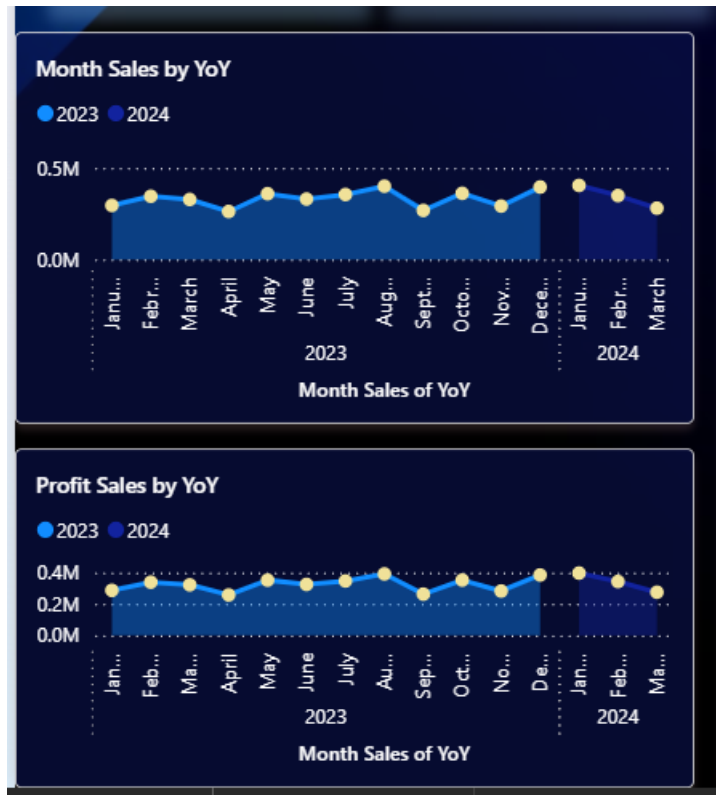


Figure 3: The Year-over-Year (YoY) analysis for 2023 and early 2024 reveals a period of high operational stability followed by a recent cooling trend. Throughout 2023, sales and profits remained consistent, peaking in August and December at approximately \$0.4M. While 2024 began with a strong January that matched the previous year's peak, the first quarter shows a gradual decline in both metrics through March. Notably, the profit trend continues to mirror sales volume almost exactly, indicating a stable but rigid margin structure that relies heavily on total revenue volume to drive bottom-line results.

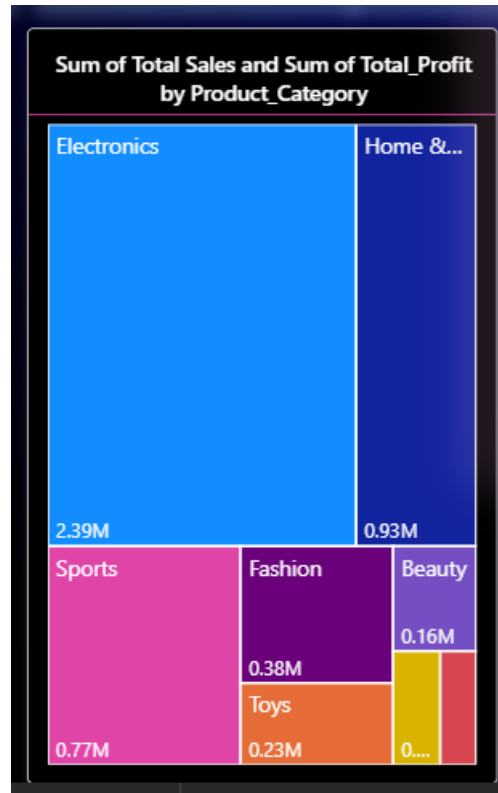


Figure 4: The tree map visualization highlights a significant dominance in the Electronics category, which accounts for the largest share of performance at \$2.39M. This is followed by Home & Kitchen (\$0.93M) and Sports (\$0.77M), which together establish the primary revenue drivers for the organization. In contrast, smaller niche categories such as Fashion (\$0.38M), Toys (\$0.23M), and Beauty (\$0.16M) contribute a combined total that is still significantly less than the Electronics sector alone. This distribution suggests a highly concentrated portfolio where overall financial health is heavily influenced by tech-related consumer trends.

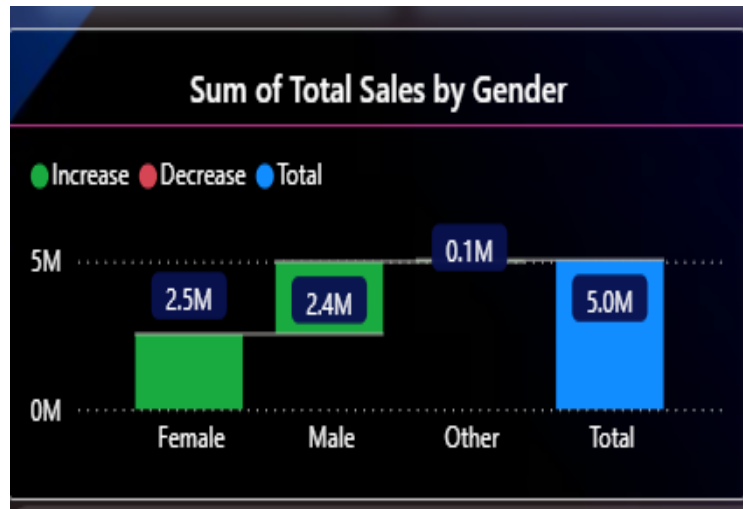


Figure 5: The waterfall chart illustrates the contribution of different demographic segments to the cumulative \$5.0M in total sales. The distribution is nearly balanced between the two primary segments, with Female customers contributing \$2.5M and Male customers accounting for \$2.4M. An additional \$0.1M is attributed to the other category, rounding out the total revenue. This indicates a highly diverse and equitable market reach, where no single gender demographic holds a dominant majority, suggesting the product line appeals broadly across the customer base.

4.3 Customer Analysis Dashboard :

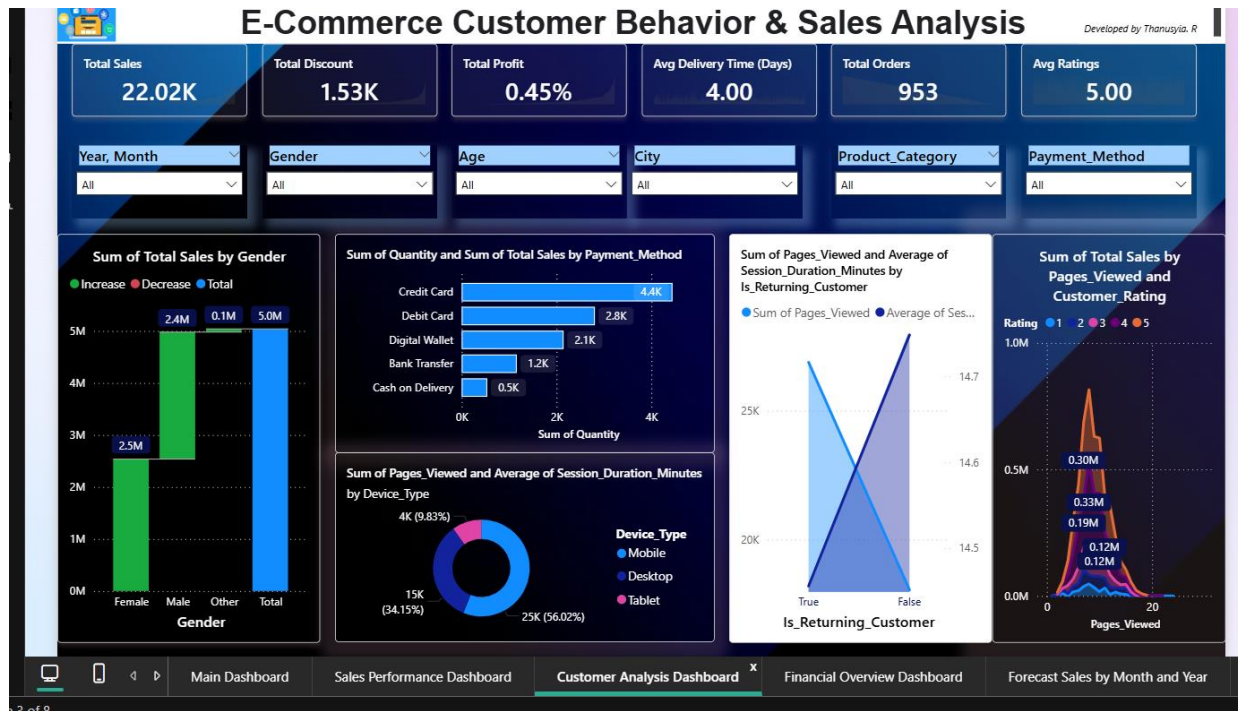


Figure 6: Introduces the Customer Analysis Dashboard; a specialized analytical environment focused on deciphering consumer habits and platform engagement. This view moves beyond purely financial metrics to investigate the "digital footprint" of the user base, correlating shopping behavior with technical accessibility and customer satisfaction.

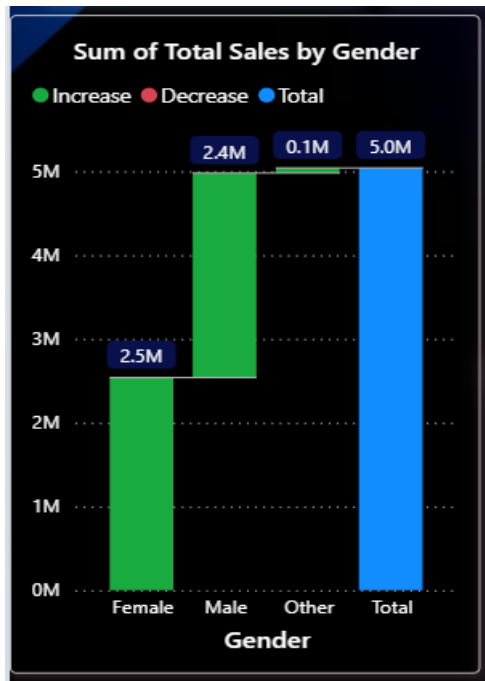


Figure7 : The waterfall chart highlights the incremental contribution of various customer segments toward the final revenue milestone of \$5.0M. Female and Male customers contribute nearly equal proportions to this total, providing \$2.5M and \$2.4M respectively, while the other segment adds the final \$0.1M. This balanced distribution underscores a highly inclusive market appeal, demonstrating that the organization’s current sales strategy is successfully reaching a broad and diverse demographic without overreliance on a single group.



Figure 8: The horizontal bar chart analyzes transaction volume by payment method, identifying Credit Card as the primary choice for customers with 4.4K units. It is followed by Debit Card (2.8K) and Digital Wallet (2.1K), highlighting a strong preference for electronic and digital payment solutions. Conversely, traditional methods like Bank Transfer (1.2K) and Cash on Delivery (0.5K) represent the smallest share of activity. This data suggests that streamlining digital checkout processes should remain a priority to align with the majority of customer purchasing behaviors.

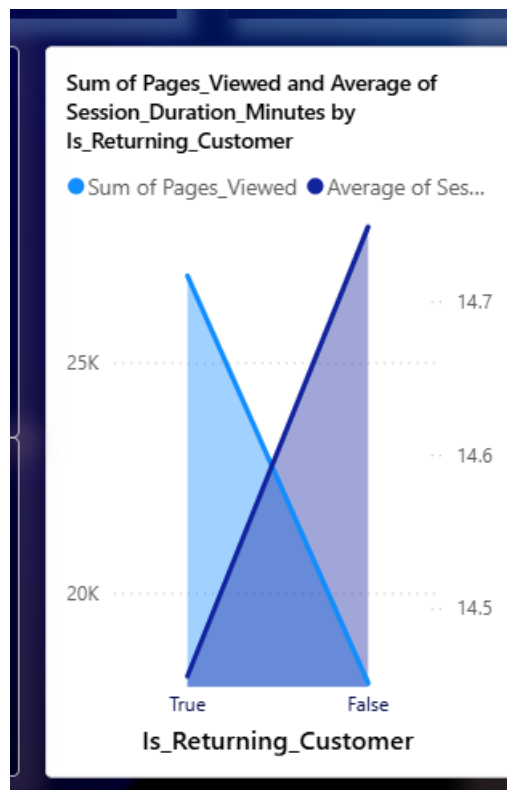


Figure 9: This dual-axis chart compares user engagement between returning and new customers, revealing an inverse relationship between volume and duration. Returning customers (True) account for a significantly higher volume of traffic, exceeding 25K pages viewed, despite spending slightly less time per session on average. Conversely, new customers (False) show lower

total page views but maintain a higher average session duration, peaking near 14.8 minutes. This suggests that while returning users navigate the site more efficiently and frequently, new users engage in deeper, more prolonged exploration during their initial visits.

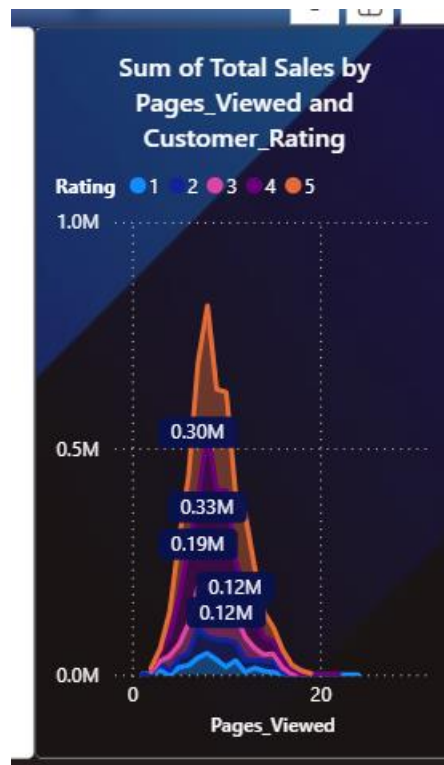


Figure 10: The ridge/area plot illustrates the correlation between customer engagement and satisfaction levels, showing that the highest concentration of revenue is generated by highly satisfied customers. Total sales peak significantly among users who provide a 5-star rating, particularly when they view between 8 and 12 pages. As the customer rating decreases, the corresponding sales volume also drops, with 1 and 2-star ratings contributing the least to the overall revenue. This distribution suggests that while a moderate level of page exploration is healthy for conversions, the ultimate driver of high-value sales is a positive user experience, as evidenced by the dominant orange peak representing top-tier satisfaction.

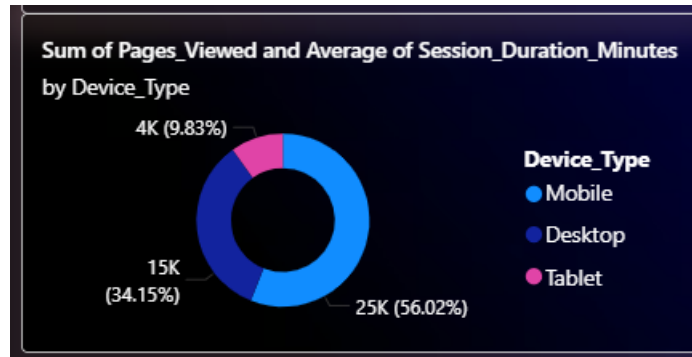


Figure 11: The chart highlights the distribution of user engagement across different hardware platforms, identifying Mobile as the primary driver of traffic with 56.02% of the total volume (25K pages viewed). Desktop users follow with a substantial 34.15% share (15K), while Tablet users represent the smallest segment at 9.83% (4K). This clear dominance of mobile activity underscores a "mobile-first" user behavior pattern, suggesting that site optimization and user interface design should prioritize the mobile experience to cater to much of the audience.

4.3 Financial Overview Dashboard



Figure 12: The Financial Overview Executive Summary, displaying the geographic distribution of revenue across Turkish provinces. This visualization establishes Istanbul as the primary economic hub, contributing \$1.4M to the total sales aggregate, while providing a comparative baseline for regional profitability.



Figure 13: illustrates the comparative growth of Monthly Sales and Monthly Profit across a 15-month period. This visualization serves as a strategic roadmap for identifying the business's seasonal rhythm and financial stability.

- **Key Finding:** The charts reveal a high degree of correlation between revenue and profit spikes, specifically peaking in August 2023 and January 2024. This synchronization indicates that the e-commerce platform maintains a healthy and consistent margin even during high-volume periods.
- **Operational Utility:** By highlighting recurring troughs in April and September, the data provides an investigative baseline for management to deploy targeted promotions during traditionally slower months.



Figure 14: utilizes a Waterfall Chart to provide a dynamic, cumulative breakdown of revenue, illustrating how each product category incrementally builds toward the total sales aggregate of 5.0M. This visualization serves as a vital investigative tool for identifying the "revenue anchors" of the business, clearly highlighting Electronics (\$2.4M) and Home & Lifestyle (\$0.9M) as the primary drivers of financial performance. By displaying the sequential contribution of diverse categories ranging from Sports to Books the chart offers a transparent view of the business's diversification and market penetration.

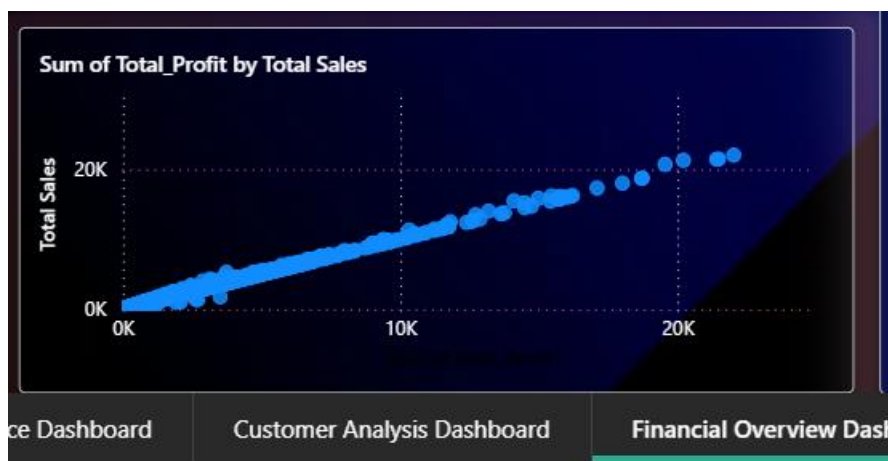


Figure 15: Sales vs. Profit Correlation Analysis (Scatter Plot)

Paragraph Summary: Figure 5 features a Scatter Plot that investigates the direct correlation between Total Sales and Total Profit, serving as a critical indicator of financial efficiency. The visual reveals a strong, linear relationship across the dataset, suggesting that as transaction volume increases, profitability scales consistently without significant margin erosion. This tight grouping of data points demonstrates a highly stable pricing and cost structure across the e-commerce platform. From an investigative standpoint, this chart allows stakeholders to quickly identify outliers; any points drifting far from the main diagonal would signal specific products or regions that are underperforming or over-performing relative to their sales volume.

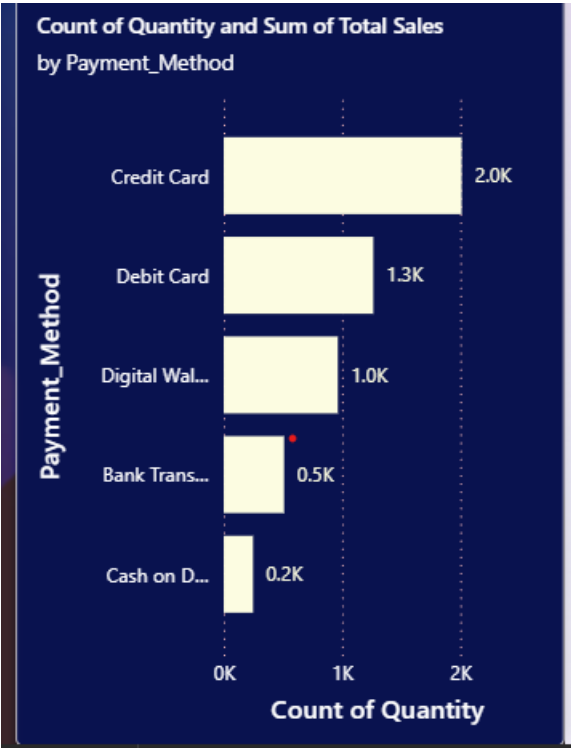


Figure 16 : Employs a horizontal bar chart to analyze the distribution of transaction volume across various payment gateways, providing critical insights into customer financial preferences within the Financial Overview Dashboard. The data reveals that Credit Cards are the primary transaction vehicle with 2.0K instances, followed by Debit Cards (1.3K) and Digital Wallets (1.0K). This ranking is an essential investigative component for understanding the business's operational liquidity and payment processing costs.



Figure 17: The organization's revenue is heavily centralized in Istanbul, which accounts for \$1.4M in total sales, significantly outperforming all other regional hubs. While daily sales exhibit high volatility with frequent spikes up to \$40K, the long-term trend remains stable. Moving into April 2024, forecasting models suggest a continuation of this baseline performance, with daily revenue expected to remain consistent with historical averages, providing a predictable foundation for Q2 planning.

Total Sales and Total Profit by Region						
Year	2023		2024		Total	
City	Average of Total Sales	Average of Total_Profit	Average of Total Sales	Average of Total_Profit	Average of Total Sales	Average of Total_Profit
▢ Konya	1,085.23	1,061.23	1,228.47	1,201.64	1,112.79	1,088.25
Beauty	215.57	213.08	229.58	217.73	218.11	213.93
Books	149.91	144.40	167.60	163.98	153.57	148.45
Electronics	3,702.02	3,609.73	4,669.25	4,579.23	3,919.65	3,827.87
Fashion	793.60	771.37	695.18	676.63	767.55	746.29
Food	174.70	170.99	90.21	90.21	157.42	154.47
Home & Garden	1,678.65	1,654.45	581.21	557.65	1,518.60	1,494.50
Sports	1,238.56	1,208.24	1,909.92	1,884.41	1,369.91	1,340.53
Toys	356.28	348.45	367.38	338.69	357.83	347.08
▢ Kayseri	1,295.71	1,247.00	1,104.43	1,083.51	1,258.50	1,215.19
Beauty	275.77	272.60	281.26	281.26	277.10	274.70
Books	85.17	81.77	60.97	59.56	81.14	78.07
Electronics	4,856.88	4,650.94	4,656.35	4,587.66	4,823.46	4,640.39
Fashion	626.50	614.85	409.45	390.36	583.09	569.95
Food	139.74	135.90	201.10	200.38	150.69	147.42
Home & Garden	1,236.97	1,212.40	972.13	848.07	1,212.14	1,178.24
Sports	1,126.16	1,069.56	1,275.07	1,275.07	1,165.58	1,123.96
Toys	377.11	366.61	328.38	317.55	365.30	354.72
▢ Izmir	951.67	930.62	1,026.46	1,005.13	967.00	945.89
Beauty	257.16	252.33	240.24	238.16	251.89	247.91
Books	92.80	90.60	90.82	88.32	92.15	89.86
Electronics	3,514.48	3,425.24	4,075.26	3,997.89	3,648.75	3,562.35
Fashion	686.10	671.15	484.82	463.81	657.35	641.53
Food	156.32	152.95	155.31	152.79	156.15	152.92
Home & Garden	1,585.75	1,562.42	1,795.27	1,758.14	1,624.12	1,598.26
Sports	1,003.89	981.10	1,185.38	1,164.41	1,028.64	1,006.10
Toys	361.61	352.43	365.56	348.29	362.23	351.77
▢ Istanbul	1,025.64	1,001.56	1,230.69	1,209.87	1,062.53	1,039.04
Beauty	265.93	258.31	251.69	247.51	263.13	256.19
Total	989.12	963.95	1,087.97	1,064.45	1,007.96	983.11

Figure 18: The comprehensive Power BI dashboard reveals an organization with a strong \$5.0M revenue foundation, driven by a balanced gender demographic and a dominant presence in the Electronics sector. Geographically, Istanbul remains the primary volume driver (\$1.4M), though city-level data indicates significant growth opportunities in Konya and Izmir, which both saw average sales increases heading into 2024. However, the business must address a cooling trend identified in Q1 2024 and the high volatility of the Home & Garden category. By leveraging the identified "satisfaction sweet-spot" of 8–12 page views and prioritizing a mobile-first user experience, the organization can stabilize recent fluctuations and decouple its profit growth from pure sales volume through the introduction of high-margin, premium product lines.

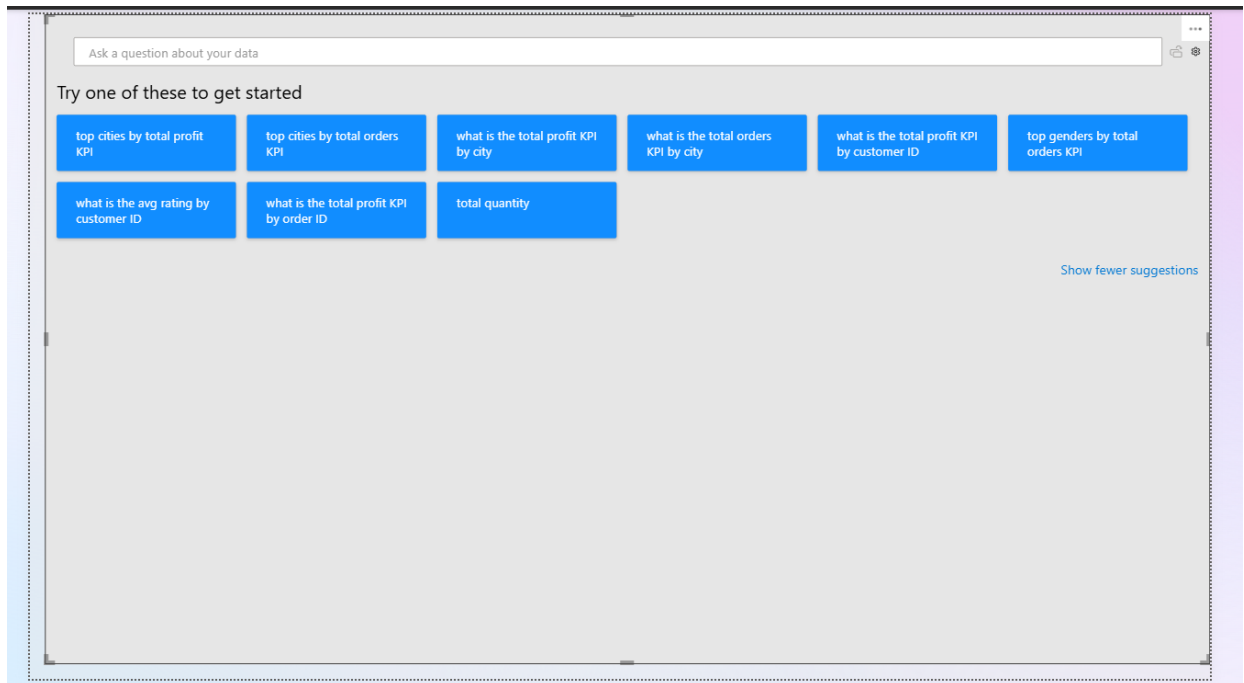


Figure 19: The implementation of the Q&A feature provides a versatile, AI-powered layer to the dashboard, enabling users to derive real-time answers through natural language queries. By offering suggested tiles for key performance indicators (KPIs) such as regional profit, order volume, and customer ratings, the interface streamlines the decision-making process. This functionality ensures that stakeholders can quickly identify top-performing cities and demographics, transforming the dashboard from a static visual tool into an interactive investigative resource that supports agile business strategies

Task 5: Insights & Analysis

The organization's performance landscape is defined by high operational stability and strong digital engagement, though it faces emerging challenges in the new fiscal year. Throughout 2023, monthly revenue remained consistent within a \$0.25M to \$0.45M range, but a distinct downward trend surfaced in Q1 2024 as performance cooled from its January peak toward a \$0.3M low in March. This revenue is heavily concentrated in the Electronics category, which dominates the portfolio at \$2.39M more than double the contribution of any other segment. Additionally, the customer base exhibits a tech-savvy, mobile-first profile, with 56% of site traffic originating from mobile devices and Credit Cards serving as the primary payment method. These trends highlight a business that is currently reliant on a single product pillar and a digital-centric audience to maintain its market position.²

The business operates on a volume-driven profit model, characterized by a near-perfect correlation between total sales and total profit. This rigid margin structure indicates that growth is currently achieved through increasing sales volume rather than through cost optimization or a shift toward high-margin products. Despite this rigidity, the brand enjoys exceptional demographic inclusivity, with a nearly balanced market reach between Female (\$2.5M) and Male (\$2.4M) segments that demonstrates a broad, universal appeal. Furthermore, data identifies a specific "satisfaction sweet-spot," where 5-star ratings and peak sales are mathematically tied to an engagement window of 8–12 page views. This suggests that the most successful user journeys are those that provide enough information to convert without overwhelming the customer, as both minimal and excessive navigation correlate with lower satisfaction and revenue.

To ensure long-term resilience and growth, the business should prioritize diversifying its product focus by reallocating marketing resources toward the Sports and Home & Kitchen categories, thereby reducing the current over-reliance on the Electronics sector. Given that 56% of traffic is mobile-based, a "mobile-first" design strategy emphasizing clear alignment and high-contrast interface elements is essential to maximize conversions for this dominant user group. To counter the emerging cooling trend in Q1 2024, the organization should implement loyalty incentives targeted at returning customers to capitalize on their high engagement and stabilize declining volumes. Finally, a strategy of margin improvement should be explored by introducing premium

product lines within the Electronics category, which would allow the business to decouple its bottom line from pure sales volume and increase overall profitability per unit.

- **Which product category performs best?**

The Electronics category is the primary revenue driver, generating \$2.39M in total sales. This is significantly higher than the next two categories, Home & Kitchen (\$0.93M) and Sports (\$0.77M). The business currently relies heavily on tech-related sales for its overall financial health.

- **Which demographic drives the most revenue?**

The market reach is remarkably inclusive and balanced. Female customers contributed \$2.5M and Male customers contributed \$2.4M, with an additional \$0.1M from other segments. This indicates a broad, universal appeal across the product lines.

- **What strategy would you recommend?** I recommend a dual-focus strategy:

Product Diversification: Increase marketing spending on Sports and Home & Kitchen to reduce the risk of over-reliance on Electronics.

Mobile UX Optimization: Continue refining the mobile interface to cater to the majority user base (56%) and implement loyalty incentives for the high-volume returning customer segment to stabilize the Q1 2024 decline.

- **What are the dominant payment methods and device types used by customers?**

Customers show a strong preference for digital engagement. Credit Cards are the leading payment method (4.4K units), followed by Debit Cards and Digital Wallets. Furthermore, 56.02% of all engagement happens via Mobile devices, highlighting the success of mobile-centric accessibility

Is there a measurable link between user engagement and customer satisfaction?

Yes. Data identifies a "satisfaction sweet-spot" where 5-star ratings and peak revenue are tied to an engagement window of 8–12 page views. While returning customers account for a significantly higher volume of traffic (25K+ page views), new customers tend to stay on the site longer per session, suggesting a deeper exploration phase during first-time visits.

Reference:

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